[INCH–POUND]
A–A–59590/22B
22 March 2013
SUPERSEDING
A–A–59590/22A
11 September 2007

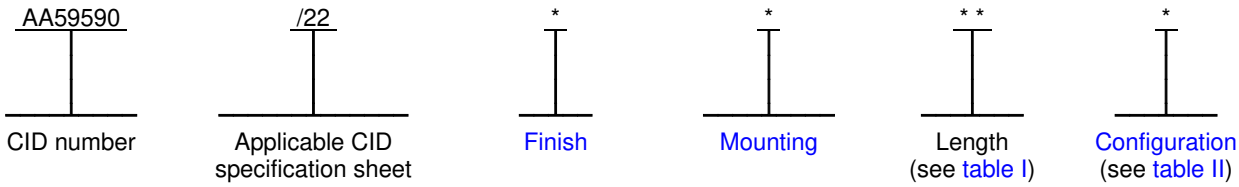
COMMERCIAL ITEM DESCRIPTION
SPECIFICATION SHEET

# HOLDER, ELECTRICAL CARD, WEDGE RETAINERS, 3 PIECE, SCREW ACTUATED DRIVE .500 X .450 INCH BODY SIZE, WITH VISUAL LOCK INDICATION

The General Services Administration has authorized the use of this commercial item description for all federal agencies.

The complete requirements for procuring electrical card holders described herein shall consist of this document and the latest issue in effect of A–A–59590.

CLASSIFICATION/PART IDENTIFICATION NUMBER (PIN). This commercial item description (CID) specification sheet uses a classification system which is included in the Part Identification Number (PIN) as shown in the following example (see notes herein).




Example: AA59590/22ET48E is the PIN for a nickel plated, 4.8 inches (122 mm) long card holder with a visual lock indication and a screw self-locking element. The card holder also features two tapped 0-80 UNF mounting holes for use with fasteners.

SALIENT CHARACTERISTICS.

Performance. Card holders shall hold the circuit card assembly firmly in its installed position and prevent loosening or movement as a result of shock and vibration. The card holder shall also provide a thermal transfer path from the circuit card assembly to the cold plate or heat sink surfaces. Card holders shall have a visual indicator to show when the assembly is in its relaxed (unlocked) state.

Interface and physical dimensions. The card holders supplied to this CID specification sheet shall be as specified herein (see figures 1, 2, 3, 4 and table I) and meet the general requirements of CID A–A–59590.

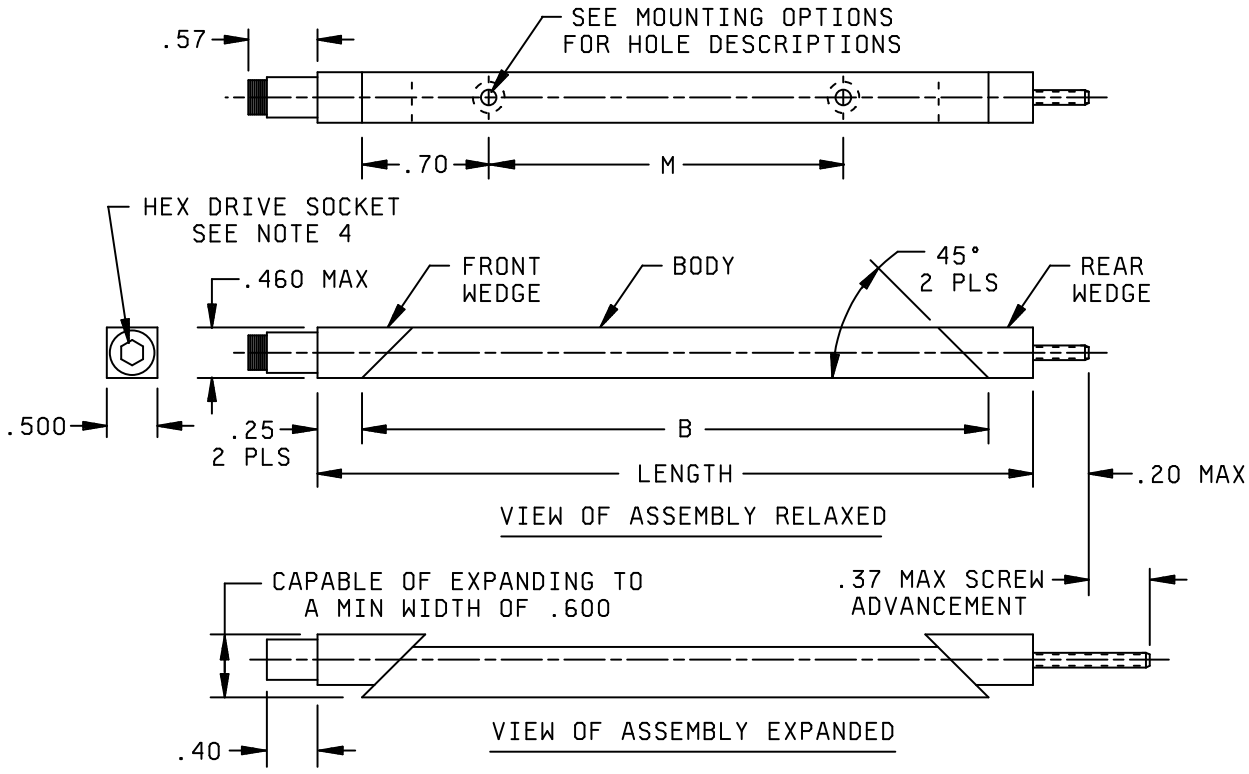
Material. Unless otherwise specified herein, the card holder materials shall be as specified in A–A–59590.

Actuating screw hex drive socket. The dimension for hex drive socket shall be .156 inch (3.96 mm) for mounting options "J", "N", "T" and "V". The dimension for hex drive socket shall be a metric dimension of 2.5 mm (.098 inch) across flats for mounting option "G".

Cold plate dimensions. Recommend cold plate slot width to accommodate the circuit card assembly with attached card holder is .525 inch (13.3 mm) plus the thickness of the printed board of the circuit card assembly (see A–A–59590).



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| Inches | mm | Inches | mm | Inches | mm | Inches | mm | Inches | mm |
|---|---|---|---|---|---|---|---|---|---|
| .20 | 5.1 | .260 | 6.60 | .27 | 6.9 | .39 | 9.9 | .70 | 17.8 |
| .25 | 6.4 | .270 | 6.86 | .325 | 8.26 | .56 | 14.2 | | |

NOTES:
1. Dimensions are in inches. Millimeters are given for general information only.
2. Unless otherwise specified, tolerances are ±.02 inch (0.51 mm) for two place decimals and ±.010 inch (0.25 mm) for three place decimals.
3. The tolerance for the mounting hole spacing is ±.005 inches (0.13 mm).
4. The across flats dimension for hex drive socket shall be .094 inch (2.39 mm) for mounting options "J", "N", "T", and "V". The across flats dimension for hex drive socket shall be a metric dimension of 2.5 mm (.098 inch) across flats for mounting option "G".

FIGURE 1. Relaxed and expanded dimensions, basic configuration.

Nominal installation torque. When card holders are used in cold plate applications, the nominal installation torque of each card holder shall be as follows: 31 inch-pounds (3.5 N-m) for assemblies using no options or option "W" and 34 to 36 inch-pounds (3.8 to 4.1 N-m) for assemblies using options "D" or "E".



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Visual lock indicator (see figure 2). Card holders shall have a visual indicator to show when the card holder assembly is in its relaxed (unlocked) state. When the card holder assembly is in the relaxed state, the end of the actuating screw shall display a red band on the side of the screw. When the actuating screw on the card holder has been tightened so that the assembly is in the expanded (locked) position, this red band shall be concealed.

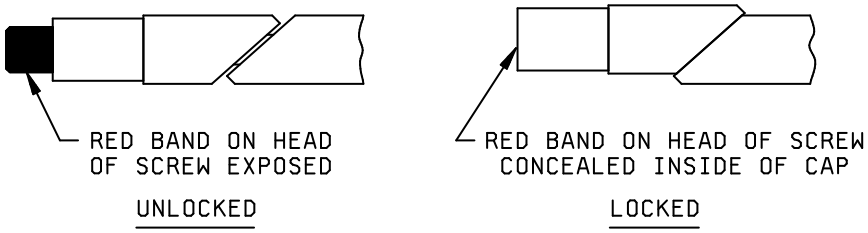



FIGURE 2. Visual lock indicator details.

Finish. The finish designator shall be as specified in A–A–59590. The finishes available for this CID specification sheet are as follows: "B" (black anodize), "C" (gold chemical film), "E" (electroless nickel), "H" (hard black anodize), or "R" (clear chemical film).

Mounting. The mounting designators shall be as specified in A–A–59590. Except when using mounting option "N", two mounting holes are required on the card holders. The mounting options available for this CID specification sheet are as follows: "J" (rivet mounting holes), "N" (no mounting holes), "T" (tapped 0-80 UNF 2B holes), "V" (tapped 2-56 UNC 2B holes), or "G" (tapped M2.5 x.45 metric holes). See figure 1 and table I for mounting hole spacing requirements. Card holders using mounting option "J" may be shipped unassembled.

Counterbore holes (for rivet mounting). The holes used for rivet mounting shall be .136 inch (3.45 mm) diameter, countersunk 100 degrees by .190/.200 inch (4.83/5.08 mm) diameter through holes with an access/clearance counterbore of .190/.200 inch (4.83/5.08 mm) diameter by .390 inch (9.91 mm) deep.

Rivets. This card holder uses rivet type B as specified in A–A–59590 when rivet mounting is used.

Length, expanded, and relaxed dimensions. The length designator shall be as specified in A–A–59590 and the lengths available for this CID specification sheet are listed in table I. The length, expanded, and relaxed dimensions shall be as specified on figure 1.

TABLE I. Assembly dimensions (see figure 1). 1/

| PIN length designator | Dimension "LENGTH" ±.02 (0.5 mm) | Dimension "B" ±.03 (0.8 mm) | Dimension "M" ±.005 (0.13 mm) |
|---|---|---|---|
| 28 | 2.8 (71 mm) | 2.30 (58.4 mm) | .90 (22.9 mm) |
| 33 | 3.3 (84 mm) | 2.80 (71.1 mm) | 1.40 (35.6 mm) |
| 38 | 3.8 (97 mm) | 3.30 (83.8 mm) | 1.90 (48.3 mm) |
| 43 | 4.3 (109 mm) | 3.80 (96.5 mm) | 2.40 (61.0 mm) |
| 48 | 4.8 (122 mm) | 4.30 (109.2 mm) | 2.90 (73.7 mm) |
| 53 | 5.3 (135 mm) | 4.80 (121.9 mm) | 3.40 (86.4 mm) |
| 58 | 5.8 (147 mm) | 5.30 (134.6 mm) | 3.90 (99.1 mm) |

1/ Dimensions are in inches. Millimeters, in parenthesis, are given for general information only.



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Configuration. The configuration of a card holder shall be as specified in table II. The details of a particular configuration consist of those on figures 1 and 2, and may include those on figures 3 and 4. The correct PIN configuration identifier for each option is listed in table II. Card holders not requiring the options described by table II shall leave the configuration position in the PIN blank.

TABLE II. Configuration.

| Configuration | Applicable figures | Hardware options |
|---|---|---|
| | 1 and 2 | No changes from basic configuration |
| W | 1, 2 and 3 | Lockwasher and flat washer |
| E | 1, 2 and 4 | Screw self-locking element |
| D | 1, 2, 3 and 4 | Lockwasher, flat washer, and screw self-locking element |

Lockwasher and flat washer (see figure 3). The use of a lockwasher and flat washer located under the screw head will provide for additional resistance to loosening of the card holder assembly from shock and vibration. Card holders requiring a lockwasher and flat washer shall include a suffix "W" in the PIN (see classification and notes).

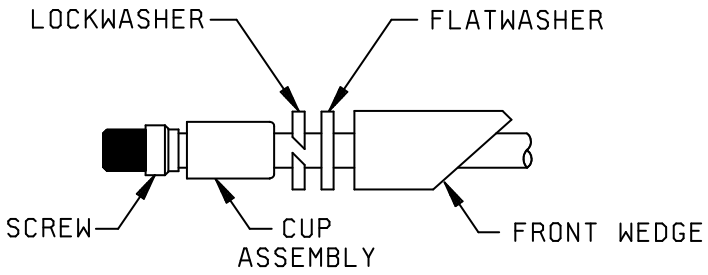



FIGURE 3. Lockwasher and flat washer details.

Screw self-locking element (see figure 4). The use of a screw self-locking element on the screw will provide a prevailing-torque for increased resistance to loosening of the card holder assembly from shock and vibration. The screw self-locking element shall be as specified in A–A–59590. Card holders requiring a screw self-locking element shall include a suffix "E" in the PIN (see classification and notes).

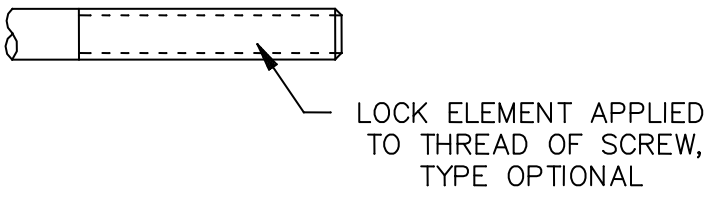



FIGURE 4. Screw self-locking element details.

Lockwasher, flat washer, and screw self-locking element. Card holders requiring a lockwasher, flat washer, and screw self-locking element option shall include a suffix "D" in the PIN (see classification and notes).



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NOTES.

PIN. The PIN should be used for Government purposes to buy commercial products to this CID specification sheet. See the classification section for PIN format example.

Source of documents.

Commercial Item Description.

A–A–59590 – Holder, Electrical Card, Wedge Retainers, 3 Piece, Screw Actuated Drive, General Requirements For.

(Copies of these documents are available online at https://assist.dla.mil/quicksearch or https://assist.dla.mil or from the Standardization Document Order Desk, 700 Robbins Avenue, Building 4D, Philadelphia, PA 19111–5094.)

Ordering data. Ordering data is as specified in A–A–59590.

Commercial products. As part of the market analysis and research effort, this CID specification sheet was coordinated with the following manufacturers of commercial products. At the time of CID specification sheet preparation and coordination, these manufacturers were known to have commercial products that would meet the requirements of this CID specification sheet. (NOTE: This information should not be considered as a list of approved manufacturers or be used to restrict procurement to only the manufacturers shown.)

| Manufacturer CAGE | Manufacturer name and address | Manufacturer contact information |
|---|---|---|
| 61081 | Calmark – A Division of Pentair Technical Products<br>7328 Trade Street<br>San Diego, CA 92121–3410 | Telephone: (858) 740–2400<br>Toll Free: (800) 854–7086<br>Facsimile: (858) 740–2430<br>E-mail: sales@calmark.com<br>URL: www.calmark.com |
| 5BG68 | American Circuit Card Retainers, Inc.<br>2310 E. Orangethorpe Avenue<br>Anaheim, CA 92806–1231 | Telephone: (714) 738–6194<br>Facsimile: (714) 446–0119<br>E-mail: sales@accrmfg.com<br>URL: http://www.accrmfg.com |

Part number supersession data. These CID specification sheet PINs supersedes the following manufacturer's part numbers as shown in table III. The CID PINs listed in table III are only for length designator "28". See table IV for CID PIN construction using other available lengths for this specification sheet. This information is being provided to assist in reducing proliferation in the Government inventory system.



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TABLE III. Commercial part number supersession data.

| PIN designator AA59590/22 | Vendor similar designator or type part number 1/ 2/ | |
|---|---|---|
| | CAGE 61081 | CAGE 5BG68 |
| BJ28 | VA255–2.80H | 3500VBA–2.80H |
| BN28 | VA255–2.80 | 3500VBA–2.80 |
| BT28 | VA255–2.80T0 | 3500VBA–2.80T0 |
| BV28 | VA255–2.80T2 | 3500VBA–2.80T2 |
| BG28 | MVA255–2.80TM2.5 | 3500MVBA–2.80TM2.5 |
| BJ28W | WVA255–2.80H | 3500VWBA–2.80H |
| BN28W | WVA255–2.80 | 3500VWBA–2.80 |
| BT28W | WVA255–2.80T0 | 3500VWBA–2.80T0 |
| BV28W | WVA255–2.80T2 | 3500VWBA–2.80T2 |
| BG28W | MWVA255–2.80TM2.5 | 3500MVWBA–2.80TM2.5 |
| BJ28E | VA255–2.80HL | 3500VBA–2.80HL |
| BN28E | VA255–2.80L | 3500VBA–2.80L |
| BT28E | VA255–2.80T0L | 3500VBA–2.80T0L |
| BV28E | VA255–2.80T2L | 3500VBA–2.80T2L |
| BG28E | MVA255–2.80TM2.5L | 3500MVBA–2.80TM2.5L |
| BJ28D | WVA255–2.80HL | 3500VWBA–2.80HL |
| BN28D | WVA255–2.80L | 3500VWBA–2.80L |
| BT28D | WVA255–2.80T0L | 3500VWBA–2.80T0L |
| BV28D | WVA255–2.80T2L | 3500VWBA–2.80T2L |
| BG28D | MWVA255–2.80TM2.5L | 3500MVWBA–2.80TM2.5L |
| CJ28 | V255–2.80H | 3500VCG–2.80H |
| CN28 | V255–2.80 | 3500VCG–2.80 |
| CT28 | V255–2.80T0 | 3500VCG–2.80T0 |
| CV28 | V255–2.80T2 | 3500VCG–2.80T2 |
| CG28 | MV255–2.80TM2.5 | 3500MVCG–2.80TM2.5 |
| CJ28W | WV255–2.80H | 3500VWCG–2.80H |
| CN28W | WV255–2.80 | 3500VWCG–2.80 |
| CT28W | WV255–2.80T0 | 3500VWCG–2.80T0 |
| CV28W | WV255–2.80T2 | 3500VWCG–2.80T2 |
| CG28W | MWV255–2.80TM2.5 | 3500MVWCG–2.80TM2.5 |

See footnotes at end of table.



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TABLE III. Commercial part number supersession data – Continued.

| PIN designator AA59590/22 | Vendor similar designator or type part number 1/ 2/ | |
|---|---|---|
| | CAGE 61081 | CAGE 5BG68 |
| CJ28E | V255–2.80HL | 3500VCG–2.80HL |
| CN28E | V255–2.80L | 3500VCG–2.80L |
| CT28E | V255–2.80T0L | 3500VCG–2.80T0L |
| CV28E | V255–2.80T2L | 3500VCG–2.80T2L |
| CG28E | MV255–2.80TM2.5L | 3500MVCG–2.80TM2.5L |
| CJ28D | WV255–2.80HL | 3500VWCG–2.80HL |
| CN28D | WV255–2.80L | 3500VWCG–2.80L |
| CT28D | WV255–2.80T0L | 3500VWCG–2.80T0L |
| CV28D | WV255–2.80T2L | 3500VWCG–2.80T2L |
| CG28D | MWV255–2.80TM2.5L | 3500MVWCG–2.80TM2.5L |
| EJ28 | VEN255–2.80H | 3500VEN–2.80H |
| EN28 | VEN255–2.80 | 3500VEN–2.80 |
| ET28 | VEN255–2.80T0 | 3500VEN–2.80T0 |
| EV28 | VEN255–2.80T2 | 3500VEN–2.80T2 |
| EG28 | MVEN255–2.80TM2.5L | 3500MVEN–2.80TM2.5 |
| EJ28W | WVEN255–2.80H | 3500VWEN–2.80H |
| EN28W | WVEN255–2.80 | 3500VWEN–2.80 |
| ET28W | WVEN255–2.80T0 | 3500VWEN–2.80T0 |
| EV28W | WVEN255–2.80T2 | 3500VWEN–2.80T2 |
| EG28W | MWVEN255–2.80TM2.5 | 3500MVWEN–2.80TM2.5 |
| EJ28E | VEN255–2.80HL | 3500VEN–2.80HL |
| EN28E | VEN255–2.80L | 3500VEN–2.80L |
| ET28E | VEN255–2.80T0L | 3500VEN–2.80T0L |
| EV28E | VEN255–2.80T2L | 3500VEN–2.80T2L |
| EG28E | MVEN255–2.80TM2.5L | 3500MVEN–2.80TM2.5L |
| EJ28D | WVEN255–2.80HL | 3500VWEN–2.80HL |
| EN28D | WVEN255–2.80L | 3500VWEN–2.80L |
| ET28D | WVEN255–2.80T0L | 3500VWEN–2.80T0L |
| EV28D | WVEN255–2.80T2L | 3500VWEN–2.80T2L |
| EG28D | MWVEN255–2.80TM2.5L | 3500MVWEN–2.80TM2.5L |

See footnotes at end of table.



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TABLE III. Commercial part number supersession data – Continued.

| PIN designator AA59590/22 | Vendor similar designator or type part number 1/ 2/ | |
|---|---|---|
| | CAGE 61081 | CAGE 5BG68 |
| HJ28 | VHA255–2.80H | 3500VBH–2.80H |
| HN28 | VHA255–2.80 | 3500VBH–2.80 |
| HT28 | VHA255–2.80T0 | 3500VBH–2.80T0 |
| HV28 | VHA255–2.80T2 | 3500VBH–2.80T2 |
| HG28 | MVHA255–2.80TM2.5L | 3500MVBH–2.80TM2.5 |
| HJ28W | WVHA255–2.80H | 3500VWBH–2.80H |
| HN28W | WVHA255–2.80 | 3500VWBH–2.80 |
| HT28W | WVHA255–2.80T0 | 3500VWBH–2.80T0 |
| HV28W | WVHA255–2.80T2 | 3500VWBH–2.80T2 |
| HG28W | MWVHA255–2.80TM2.5L | 3500MVWBH–2.80TM2.5 |
| HJ28E | VHA255–2.80HL | 3500VBH–2.80HL |
| HN28E | VHA255–2.80L | 3500VBH–2.80L |
| HT28E | VHA255–2.80T0L | 3500VBH–2.80T0L |
| HV28E | VHA255–2.80T2L | 3500VBH–2.80T2L |
| HG28E | VMHA255–2.80TM2.5L | 3500MVBH–2.80TM2.5 |
| HJ28D | WVHA255–2.80HL | 3500VWBH–2.80HL |
| HN28D | WVHA255–2.80L | 3500VWBH–2.80L |
| HT28D | WVHA255–2.80T0L | 3500VWBH–2.80T0L |
| HV28D | WVHA255–2.80T2LR | 3500VWBH–2.80T2L |
| HG28D | MWVHA255–2.80TM2.5L | 3500MVWBH–2.80TM2.5L |
| RJ28 | VR255–2.80H | 3500VCC–2.80H |
| RN28 | VR255–2.80 | 3500VCC–2.80 |
| RT28 | VR255–2.80T0 | 3500VCC–2.80T0 |
| RV28 | VR255–2.80T2 | 3500VCC–2.80T2 |
| RG28 | MVR255–2.80TM2.5L | 3500MVCC–2.80TM2.5 |
| RJ28W | WVR255–2.80H | 3500VWCC–2.80H |
| RN28W | WVR255–2.80 | 3500VWCC–2.80 |
| RT28W | WVR255–2.80T0 | 3500VWCC–2.80T0 |
| RV28W | WVR255–2.80T2 | 3500VWCC–2.80T2 |
| RG28W | MWVR255–2.80TM2.5L | 3500MVWCC–2.80TM2.5 |

See footnotes at end of table.



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TABLE III. Commercial part number supersession data – Continued.

| PIN designator AA59590/22 | Vendor similar designator or type part number 1/ 2/ | |
|---|---|---|
| | CAGE 61081 | CAGE 5BG68 |
| RJ28E | VR255–2.80HL | 3500VCC–2.80HL |
| RN28E | VR255–2.80L | 3500VCC–2.80L |
| RT28E | VR255–2.80T0L | 3500VCC–2.80T0L |
| RV28E | VR255–2.80T2L | 3500VCC–2.80T2L |
| RG28E | VMR255–2.80TM2.5L | 3500MVCC–2.80TM2.5L |
| RJ28D | WVR255–2.80HL | 3500VWCC–2.80HL |
| RN28D | WVR255–2.80L | 3500VWCC–2.80L |
| RT28D | WVR255–2.80T0L | 3500VWCC–2.80T0L |
| RV28D | WVR255–2.80T2L | 3500VWCC–2.80T2L |
| RG28D | MWVR255–2.80TM2.5L | 3500MVWCC–2.80TM2.5L |

1/ The CID PINs listed are only for length designator "28".
2/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.

TABLE IV. Example of PIN with available length designators.

| PIN designator AA59590/22 | Vendor similar designator or type part number 1/ 2/ | |
|---|---|---|
| | CAGE 61081 | CAGE 5BG68 |
| ET28E | VEN255–2.80T0L | 3500VEN–2.80T0L |
| ET38E | VEN255–3.80T0L | 3500VEN–3.80T0L |
| ET48E | VEN255–4.80T0L | 3500VEN–5.80T0L |
| ET58E | VEN255–5.80T0L | 3500VEN–5.80T0L |

1/ The manufacturer's part number shall not be used for procurement to the requirements of this CID specification sheet. At the time of preparation of this CID specification sheet, the aforementioned commercial products were reviewed and could be replaced by the CID PIN shown. For actual part marking requirements, see the marking paragraph in A–A–59590.
2/ Other lengths are available on request.



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MILITARY INTERESTS:

Custodians:
Army – CR
Navy – EC
Air Force – 85
DLA – CC

Review activity:
Air Force – 99

CIVIL AGENCY COORDINATING ACTIVITY:

GSA – FAS

Preparing activity:
DLA – CC

Project 5998–2012–022

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